

3-DIMENSIONAL GEOMETRY

1.	Direction cosines of the line $\frac{x-1}{2} = \frac{1-y}{3} = \frac{2z-1}{12}$ are : (a) $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$ (b) $\frac{2}{\sqrt{157}}, -\frac{3}{\sqrt{157}}, \frac{12}{\sqrt{157}}$ (c) $\frac{2}{7}, -\frac{3}{7}, -\frac{6}{7}$ (d) $\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}$
2.	Assertion (A) : Equation of a line passing through the points (1, 2, 3) and (3, -1, 3) is $\frac{x-3}{2} = \frac{y+1}{3} = \frac{z-3}{0}$. Reason (R) : Equation of a line passing through points (x_1, y_1, z_1), (x_2, y_2, z_2) is given by $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$.
3.	Check whether the lines given by equations $x = 2\lambda + 2$, $y = 7\lambda + 1$, $z = -3\lambda - 3$ and $x = -\mu - 2$, $y = 2\mu + 8$, $z = 4\mu + 5$ are perpendicular to each other or not.
4.	Find the vector and the Cartesian equations of a line passing through the point (1, 2, -4) and parallel to the line joining the points A(3, 3, -5) and B(1, 0, -11). Hence, find the distance between the two lines.
5.	Find the equations of the line passing through the points A(1, 2, 3) and B(3, 5, 9). Hence, find the coordinates of the points on this line which are at a distance of 14 units from point B.
6.	Equation of line passing through origin and making 30°, 60° and 90° with x, y, z axes respectively is (A) $\frac{2x}{\sqrt{3}} = \frac{y}{2} = \frac{z}{0}$ (B) $\frac{2x}{\sqrt{3}} = \frac{2y}{1} = \frac{z}{0}$ (C) $2x = \frac{2y}{\sqrt{3}} = \frac{z}{1}$ (D) $\frac{2x}{\sqrt{3}} = \frac{2y}{1} = \frac{z}{1}$
7.	Assertion (A) : If a line makes angles α, β, γ with positive direction of the coordinate axes, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$. Reason (R) : The sum of squares of the direction cosines of a line is 1.
8.	If the equation of a line is $x = ay + b$, $z = cy + d$, then find the direction ratios of the line and a point on the line.

9.	Find the equations of the diagonals of the parallelogram PQRS whose vertices are P(4, 2, -6), Q(5, -3, 1), R(12, 4, 5) and S(11, 9, -2). Use these equations to find the point of intersection of diagonals.
10.	A line l passes through point $(-1, 3, -2)$ and is perpendicular to both the lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$. Find the vector equation of the line l . Hence, obtain its distance from origin.
11.	The equation of a line passing through point $(2, -1, 0)$ and parallel to the line $\frac{x}{1} = \frac{y-1}{2} = \frac{2-z}{2}$ is : (a) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{2}$ (b) $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z}{2}$ (c) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{-2}$ (d) $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z}{-2}$
12.	Assertion (A) : A line through the points $(4, 7, 8)$ and $(2, 3, 4)$ is parallel to a line through the points $(-1, -2, 1)$ and $(1, 2, 5)$. Reason (R): Lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ are parallel if $\vec{b}_1 \cdot \vec{b}_2 = 0$.
13.	Find the direction cosines of the line whose Cartesian equations are $5x - 3 = 15y + 7 = 3 - 10z$.
14.	Find the value of b so that the lines $\frac{x-1}{2} = \frac{y-b}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$ are intersecting lines. Also, find the point of intersection of these given lines.
15.	Find the equations of all the sides of the parallelogram ABCD whose vertices are A(4, 7, 8), B(2, 3, 4), C(-1, -2, 1) and D(1, 2, 5). Also, find the coordinates of the foot of the perpendicular from A to CD.
16.	Distance of the point (p, q, r) from y-axis is : (a) q (b) q (c) $q + r$ (d) $\sqrt{p^2 + r^2}$

17.	If the direction cosines of a line are $\left(\frac{1}{a}, \frac{1}{a}, \frac{1}{a}\right)$, then : (a) $0 < a < 1$ (b) $a > 2$ (c) $a > 0$ (d) $a = \pm \sqrt{3}$
18.	<i>Assertion (A) :</i> The lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ are perpendicular, when $\vec{b}_1 \cdot \vec{b}_2 = 0$. <i>Reason (R) :</i> The angle θ between the lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is given by $\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{ \vec{b}_1 \vec{b}_2 }$
19.	Find the vector equation of the line passing through the point (2, 1, 3) and perpendicular to both the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}; \quad \frac{x}{-3} = \frac{y}{2} = \frac{z}{5}.$
20.	The equations of a line are $5x - 3 = 15y + 7 = 3 - 10z$. Write the direction cosines of the line and find the coordinates of a point through which it passes.
21.	Show that the following lines do not intersect each other : $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}; \quad \frac{x+2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$
22.	Find the angle between the lines $2x = 3y = -z \text{ and } 6x = -y = -4z.$
23.	The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is (a) 0° (b) 30° (c) 45° (d) 90°
24.	If a line makes angles of 90°, 135° and 45° with the x, y and z axes respectively, then its direction cosines are (a) $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$ (c) $\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}$ (d) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

25.	Find the angle between the following two lines : $\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k}) ;$ $\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$
26.	Find the distance between the lines : $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k}) ;$ $\vec{r} = (3\hat{i} + 3\hat{j} - 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$
27.	Find the coordinates of the foot of the perpendicular drawn from the point P(0, 2, 3) to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$.
28.	The direction ratios of the line $\frac{x-1}{3} = \frac{2-y}{1} = \frac{3z}{2}$ are : (A) 3, 1, 2 (B) 4, 3, 2 (C) 9, -3, 2 (D) 9, 3, 2
29.	The Cartesian equation of the line passing through the point (1, -3, 2) and parallel to the line $\vec{r} = 2\hat{i} - \hat{k} + \lambda(\hat{i} + \hat{j} + 2\hat{k})$ is : (A) $\frac{x-1}{2} = \frac{y+3}{0} = \frac{z-2}{-1}$ (B) $\frac{x+1}{1} = \frac{y-3}{1} = \frac{z+2}{2}$ (C) $\frac{x+1}{2} = \frac{y-3}{0} = \frac{z+2}{-1}$ (D) $\frac{x-1}{1} = \frac{y+3}{1} = \frac{2-z}{-2}$
30.	Find the angle between the lines $\frac{5-x}{-7} = \frac{y+2}{-5} = \frac{z}{1} \text{ and } \frac{x}{1} = \frac{y}{2} = \frac{z}{3}.$

31.	If the direction cosines of a line are $\sqrt{3}k, \sqrt{3}k, \sqrt{3}k$, then the value of k is : (A) ± 1 (B) $\pm \sqrt{3}$ (C) ± 3 (D) $\pm \frac{1}{3}$
32.	Find the shortest distance between the lines $\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.
33.	Find the point of intersection of the lines $\vec{r} = \hat{i} - \hat{j} + 6\hat{k} + \lambda(3\hat{i} - \hat{k})$, and $\vec{r} = -3\hat{j} + 3\hat{k} + \mu(\hat{i} + 2\hat{j} - \hat{k})$. Also, find the vector equation of the line passing through the point of intersection of the given lines and perpendicular to both the lines.
34.	The vector equation of a line which passes through the point $(2, -4, 5)$ and is parallel to the line $\frac{x+3}{3} = \frac{4-y}{2} = \frac{z+8}{6}$ is : (a) $\vec{r} = (-2\hat{i} + 4\hat{j} - 5\hat{k}) + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ (b) $\vec{r} = (2\hat{i} - 4\hat{j} + 5\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$ (c) $\vec{r} = (2\hat{i} - 4\hat{j} + 5\hat{k}) + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ (d) $\vec{r} = (-2\hat{i} + 4\hat{j} - 5\hat{k}) + \lambda(3\hat{i} - 2\hat{j} - 6\hat{k})$
35.	<i>Assertion (A) :</i> Quadrilateral formed by vertices $A(0, 0, 0)$, $B(3, 4, 5)$, $C(8, 8, 8)$ and $D(5, 4, 3)$ is a rhombus. <i>Reason (R) :</i> ABCD is a rhombus if $AB = BC = CD = DA$, $AC \neq BD$.
36.	If a line makes angles α, β and γ with x-axis, y-axis and z-axis respectively, then prove that $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.
37.	Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Also find their point of intersection.

38.	Find the shortest distance between the pair of lines $\frac{x-1}{2} = \frac{y+1}{3} = z$ and $\frac{x+1}{5} = \frac{y-2}{1}; z = 2$.
39.	If the direction cosines of a line are $\sqrt{3} k, \sqrt{3} k, \sqrt{3} k$, then the value of k is : (A) ± 1 (B) $\pm \sqrt{3}$ (C) ± 3 (D) $\pm \frac{1}{3}$
40.	The distance of point P(a, b, c) from y-axis is : (A) b (B) b^2 (C) $\sqrt{a^2 + c^2}$ (D) $a^2 + c^2$
41.	<i>Assertion (A) :</i> A line in space cannot be drawn perpendicular to x, y and z axes simultaneously. <i>Reason (R) :</i> For any line making angles, α, β, γ with the positive directions of x, y and z axes respectively, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.
42.	The image of point P(x, y, z) with respect to line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is P' (1, 0, 7). Find the coordinates of point P.
43.	The coordinates of the foot of the perpendicular drawn from the point (0, 1, 2) on the x-axis are given by : (A) (1, 0, 0) (B) (2, 0, 0) (C) $(\sqrt{5}, 0, 0)$ (D) (0, 0, 0)
44.	Direction ratios of a vector parallel to line $\frac{x-1}{2} = -y = \frac{2z+1}{6}$ are : (A) 2, -1, 6 (B) 2, 1, 6 (C) 2, 1, 3 (D) 2, -1, 3

45.	If a line makes an angle of 30° with the positive direction of x-axis, 120° with the positive direction of y-axis, then the angle which it makes with the positive direction of z-axis is : (A) 90° (B) 120° (C) 60° (D) 0°
46.	Find the value of p for which the lines $\vec{r} = \lambda \hat{i} + (2\lambda + 1)\hat{j} + (3\lambda + 2)\hat{k}$ and $\vec{r} = \hat{i} - 3\mu\hat{j} + (p\mu + 7)\hat{k}$ are perpendicular to each other and also intersect. Also, find the point of intersection of the given lines.
47.	If a line makes an angle of $\frac{\pi}{4}$ with the positive directions of both x-axis and z-axis, then the angle which it makes with the positive direction of y-axis is : (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π
48.	If α, β and γ are the angles which a line makes with positive directions of x, y and z axes respectively, then which of the following is not true ? (A) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ (B) $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$ (C) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$ (D) $\cos \alpha + \cos \beta + \cos \gamma = 1$
49.	Find the equation of the line passing through the point of intersection of the lines $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and $\frac{x-1}{0} = \frac{y}{-3} = \frac{z-7}{2}$ and perpendicular to these given lines.
50.	Two vertices of the parallelogram ABCD are given as A(-1, 2, 1) and B(1, -2, 5). If the equation of the line passing through C and D is $\frac{x-4}{1} = \frac{y+7}{-2} = \frac{z-8}{2}$, then find the distance between sides AB and CD. Hence, find the area of parallelogram ABCD.

51.	The vector equation of a line passing through the point $(1, -1, 0)$ and parallel to Y-axis is : (A) $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} - \hat{j})$ (B) $\vec{r} = \hat{i} - \hat{j} + \lambda\hat{j}$ (C) $\vec{r} = \hat{i} - \hat{j} + \lambda\hat{k}$ (D) $\vec{r} = \lambda\hat{j}$
52.	The lines $\frac{1-x}{2} = \frac{y-1}{3} = \frac{z}{1}$ and $\frac{2x-3}{2p} = \frac{y}{-1} = \frac{z-4}{7}$ are perpendicular to each other for p equal to : (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 2 (D) 3
53.	Find the distance between the line $\frac{x}{2} = \frac{2y-6}{4} = \frac{1-z}{-1}$ and another line parallel to it passing through the point $(4, 0, -5)$.
54.	If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-7}$ are perpendicular to each other, find the value of k and hence write the vector equation of a line perpendicular to these two lines and passing through the point $(3, -4, 7)$.
55.	The angle which the line $\frac{x}{1} = \frac{y}{-1} = \frac{z}{0}$ makes with the positive direction of Y-axis is : (A) $\frac{5\pi}{6}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$
56.	The Cartesian equation of the line passing through the point $(1, -3, 2)$ and parallel to the line : $\vec{r} = (2 + \lambda)\hat{i} + \lambda\hat{j} + (2\lambda - 1)\hat{k}$ is (A) $\frac{x-1}{2} = \frac{y+3}{0} = \frac{z-2}{-1}$ (B) $\frac{x+1}{1} = \frac{y-3}{1} = \frac{z+2}{2}$ (C) $\frac{x+1}{2} = \frac{y-3}{0} = \frac{z+2}{-1}$ (D) $\frac{x-1}{1} = \frac{y+3}{1} = \frac{z-2}{2}$

57.	Find the vector equation of the line passing through the point (2, 3, -5) and making equal angles with the co-ordinate axes.
58.	Find the co-ordinates of the foot of the perpendicular drawn from the point (2, 3, -8) to the line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$. Also, find the perpendicular distance of the given point from the line.
59.	Find the shortest distance between the lines L_1 & L_2 given below : L_1 : The line passing through (2, -1, 1) and parallel to $\frac{x}{1} = \frac{y}{1} = \frac{z}{3}$ L_2 : $\vec{r} = \hat{i} + (2\mu + 1)\hat{j} - (\mu + 2)\hat{k}$.
60.	Verify that lines given by $\vec{r} = (1 - \lambda)\hat{i} + (\lambda - 2)\hat{j} + (3 - 2\lambda)\hat{k}$ and $\vec{r} = (\mu + 1)\hat{i} + (2\mu - 1)\hat{j} - (2\mu + 1)\hat{k}$ are skew lines. Hence, find shortest distance between the lines.
61.	Find the image A' of the point A(1, 6, 3) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Also, find the equation of the line joining A and A'.
62.	Find a point P on the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ such that its distance from point Q(2, 4, -1) is 7 units. Also, find the equation of line joining P and Q.
63.	The equation of a line parallel to the vector $3\hat{i} + \hat{j} + 2\hat{k}$ and passing through the point (4, -3, 7) is : (A) $x = 4t + 3, y = -3t + 1, z = 7t + 2$ (B) $x = 3t + 4, y = t + 3, z = 2t + 7$ (C) $x = 3t + 4, y = t - 3, z = 2t + 7$ (D) $x = 3t + 4, y = -t + 3, z = 2t + 7$
64.	The line $x = 1 + 5\mu, y = -5 + \mu, z = -6 - 3\mu$ passes through which of the following point ? (A) (1, -5, 6) (B) (1, 5, 6) (C) (1, -5, -6) (D) (-1, -5, 6)

65.	Find the shortest distance between the lines : $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3} \text{ and }$ $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}.$
66.	Find the image A' of the point $A(2, 1, 2)$ in the line $l : \vec{r} = 4\hat{i} + 2\hat{j} + 2\hat{k} + \lambda (\hat{i} - \hat{j} - \hat{k})$. Also, find the equation of line joining AA'. Find the foot of perpendicular from point A on the line l.
67.	The coordinates of the foot of the perpendicular drawn from the point $A(-2, 3, 5)$ on the y-axis is : (A) $(0, 0, 5)$ (B) $(0, 3, 0)$ (C) $(-2, 0, 5)$ (D) $(-2, 0, 0)$
68.	Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.
69.	Find the equation of a line in vector and cartesian form which passes through the point $(1, 2, -4)$ and is perpendicular to the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$, and $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k}).$
70.	Find the foot of the perpendicular drawn from the point $(1, 1, 4)$ on the line $\frac{x+2}{5} = \frac{y+1}{2} = \frac{-z+4}{-3}$.
71.	Find the point on the line $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-4}{3}$ at a distance of $2\sqrt{2}$ units from the point $(-1, -1, 2)$.

72.	Find the value of λ if the following lines are perpendicular to each other : $l_1: \frac{1-x}{-3} = \frac{3y-2}{2\lambda} = \frac{z-3}{3}$ $l_2: \frac{x-1}{3\lambda} = \frac{1-y}{1} = \frac{2z-5}{3}$
73.	Find the distance of the point $P(2, 4, -1)$ from the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$.
74.	Let the position vectors of the points A, B and C be $3\hat{i} - \hat{j} - 2\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$ and $\hat{i} + 5\hat{j} + 3\hat{k}$ respectively. Find the vector and cartesian equations of the line passing through A and parallel to line BC.
75.	Find the foot of the perpendicular drawn from point $(2, -1, 5)$ to the line $\frac{x-11}{10} = \frac{y+2}{-4} = \frac{z+8}{-11}$. Also, find the length of the perpendicular.
76.	Find the shortest distance between the lines : $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda (\hat{i} - 2\hat{j} + 3\hat{k})$ $\vec{r} = (\hat{i} + 4\hat{k}) + \mu (3\hat{i} - 6\hat{j} + 9\hat{k}).$
77.	Find the point Q on the line $\frac{2x+4}{6} = \frac{y+1}{2} = \frac{-2z+6}{-4}$ at a distance of $3\sqrt{2}$ from the point $P(1, 2, 3)$.
78.	Find the image of the point $(-1, 5, 2)$ in the line $\frac{2x-4}{2} = \frac{y}{2} = \frac{2-z}{3}$. Find the length of the line segment joining the points (given point and the image point).

CASE BASED QUESTIONS

79.

An engineer is designing a new metro rail network in a city.



Initially, two metro lines, Line A and Line B, each consisting of multiple stations are designed. The track for Line A is represented by

$$l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{4}, \text{ while the track for Line B is represented by } l_2 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z+2}{-3}.$$

Based on the above information, answer the following questions :

- (i) Find whether the two metro tracks are parallel. 1
- (ii) Solar panels are to be installed on the rooftop of the metro stations. Determine the equation of the line representing the placement of solar panels on the rooftop of Line A's stations, given that panels are to be positioned parallel to Line A's track (l_1) and pass through the point $(1, -2, -3)$. 1
- (iii) (a) To connect the stations, a pedestrian pathway perpendicular to the two metro lines is to be constructed which passes through point $(3, 2, 1)$. Determine the equation of the pedestrian walkway. 2

OR

- (iii) (b) Find the shortest distance between Line A and Line B. 2